

The background of the slide is a close-up, high-contrast image of fire. Bright orange and yellow flames are visible, with a dark, almost black, central area where the fire is most intense. The flames appear to be rising and spreading, creating a sense of movement and heat. The overall effect is dramatic and visually striking.

# Writing with fire

**Thermal decomposition of nitrates**



# IMPORTANT MESSAGE from Dr. Dragon...

Fire very dangerous. I am a dragon I would know. Be safe.



Make sure during the experiment...

1. You're wearing **goggles**
2. You're **hair is tied back**
3. You **listen** to the instructions and regulations
4. Only **teachers** or **Sixth Formers** can use the **fire**



Time for the fun part!

## Method:

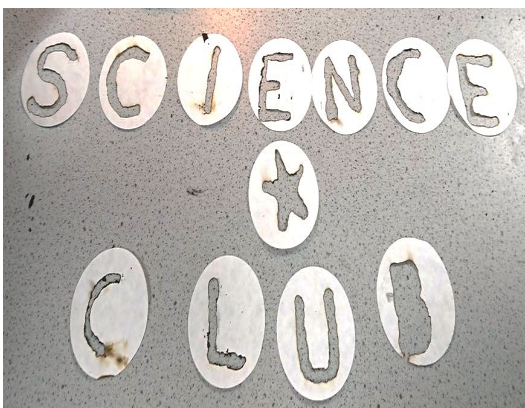
### Children:

1. Using a small paintbrush and the magic ink to write a message on the absorbent paper. Use joined up writing! Design the message so that the sheet of paper will remain in one piece after the message has burnt through.
2. Try not to make your writing too thick!
3. Mark the start of it with a light pencil mark.
4. Thoroughly dry the message using a hot-air blower, or by holding the paper well above a Bunsen flame.



### Teacher/ Sixth Former:

5. Apply a glowing splint to the start of the message until the treated paper starts to glow and char.
6. Remove the splint and watch as the glow and charring work their way along the message, leaving the untreated paper untouched.



Do you know which three  
things we need to make fire?

# The Fire Triangle

THE FIRE TRIANGLE

Energy

Oxygen

Fuel

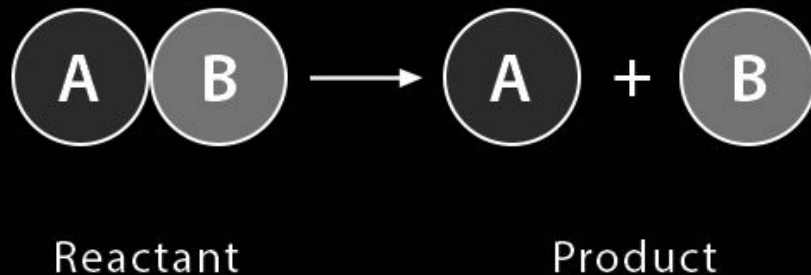


What does decomposition mean?



# The Decomposition of Nitrates

The decomposition of nitrates



But what does decomposition have to do with this?

# The Decomposition of Nitrates

The decomposition of nitrates

Sodium nitrate decomposes to give sodium nitrite [sodium nitrate(III)] and oxygen, and it is the oxygen released which helps to promote the burning process and produce the glow and charring:

